

Stacked Polyphase Bridge Converters for High-Voltage Electric Traction Drives: A Review of Topology, Modulation, Control, Semiconductor Scaling, and Future Directions

Enes Ayaz, *Student Member, IEEE*, Shahriar Sarmast Ghahfarokhi, *Student Member, IEEE*, Staffan Norrga, *Member, IEEE*, Hans-Peter Nee, *Fellow, IEEE*

Abstract—Increasing dc-link voltages in electric traction drives intensify semiconductor voltage stress and insulation requirements. Stacked polyphase bridge converters partition the dc-link voltage among series-connected three-phase bridges, each supplying an electrically separated machine winding group. This architecture enables lower-voltage semiconductor devices and modular converter-machine integration, but introduces coupled challenges in modulation, voltage balancing, insulation, and fault management. This paper reviews stacked polyphase bridge converters for high-voltage electric traction, bringing together literature on topology evolution, operating principles, machine integration, control, semiconductor scaling, and fault-tolerant reconfiguration. The review examines carrier-interleaving effects on dc-link ripple, conditions for switching-induced common-mode voltage cancellation, and dc-link balancing stability under centralized, common-duty-ratio, and distributed control. It also relates cell count to semiconductor voltage class and the associated converter and machine requirements. A mapping for a 1200-V dc link illustrates how increasing cell count provides access to lower-voltage device classes, while adding sensing, gate-drive, winding, and insulation requirements. The synthesis indicates that semiconductor-level gains alone are insufficient to establish a system-level advantage, and that the preferred cell count depends on the application and operating mission. Key research priorities are joint battery, converter, and machine design, together with validation against vehicle-level energy, cost, and reliability targets. These findings provide a basis for assessing where stacked polyphase bridge architectures can offer practical benefits in future traction drives.

Index Terms—Electric vehicles, integrated motor drives, modular converters, multiphase machines, stacked polyphase bridge (SPB) converters.

I. INTRODUCTION

Electrification is pushing traction-drive propulsion and fast-charging power toward several hundred kilowatts, while system-level performance requirements continue to rise [1]. Increasing the dc-link voltage has emerged as a primary design lever for meeting both demands. On the propulsion side, a higher dc-link voltage improves efficiency and power density [2]. On the battery and charging side, it reduces the current required for a given power level, lowering cable conduction loss and conductor cross-section while enabling higher charging power under a given current limit [3].

(Corresponding author: Enes Ayaz.)

E. Ayaz, S. Sarmast Ghahfarokhi, S. Norrga, and H.-P. Nee are with the Division of Electric Power and Energy Systems, KTH Royal Institute of Technology, SE-100 44 Stockholm, Sweden (e-mail: enesa@kth.se).

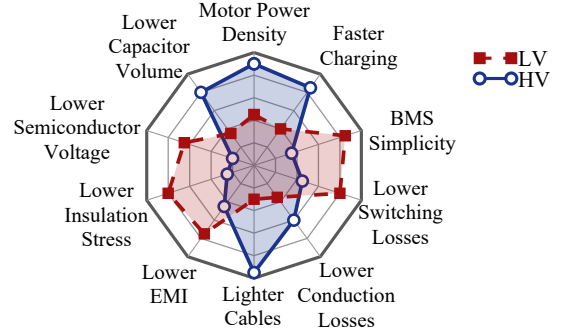


Fig. 1. Conceptual illustration of the qualitative system-level tradeoffs between a low-voltage and a high-voltage traction architecture.

This trend, however, is not without consequence at the converter level. A higher dc-link voltage also brings more severe insulation-coordination and electromagnetic-compatibility (EMC) demands, together with added thermal and reliability stress on the power stage [2], [3]. Fig. 1 summarizes this qualitative tradeoff between a conventional low-voltage and a higher-voltage traction architecture. Raising the dc-link voltage therefore does not amount to a simple rescaling of an existing inverter design; it reshapes the architecture-level design space of the complete electric drive.

A. Literature and Industry Trends Toward Higher DC-Link Voltage

The trend motivated above is not merely conceptual: it is already established in academic prototypes, industrial inverter platforms, and production vehicles, as summarized in Table I. Reported academic demonstrators range from 400-V to 800-V and 1.2-kV converters, with still higher levels investigated for high-power passenger and heavy-duty applications [3], while industrial platforms increasingly target 800-V operation as a baseline and production vehicles show a comparable shift from established 400-V architectures to a growing number of 800-V and higher-voltage platforms [4], [5]. This industry-wide direction, rather than an isolated research interest, motivates the semiconductor-level analysis that follows since the rising dc-link voltages translate directly into a semiconductor-level bottleneck.

TABLE I

REPRESENTATIVE VOLTAGE AND POWER LEVELS IN PUBLISHED TRACTION-INVERTER STUDIES, INDUSTRIAL AND REFERENCE PLATFORMS, AND PRODUCTION BATTERY-ELECTRIC VEHICLES.

Published studies / demonstrators	Industrial / reference platforms	Production vehicles
Bertelshofer et al.: 400 V, 150 kW [6]	Bosch Gen. 4:400/800 V, 250/350 kW [7]	BMW i4 M50: 400 V, 400 kW [8]
Taha et al.: 800 V, 100 kW [9]	Wolfspeed reference design: 800 V, 300 kW [10]	Hyundai IONIQ 5 N: 800 V, 478 kW [11]
Al-Hmoud et al.: 1 kV, 200 kW [12]	Infineon eCAV platform: 800 V, 250 kW [13]	Volvo EX90 Performance: 800 V, 500 kW [14]
Wang et al.: 1.2 kV, 100 kW [15]	AVL PI850e: 800 V, 250 kW [16]	Porsche Taycan Turbo S: 800 V, 700 kW [17]
Ayaz et al.: 1.2 kV, 250 kW [18]	Punch IV5: 800 V, 600 kW [19]	Lucid Air Dream Performance: 900 V, 828 kW [20]

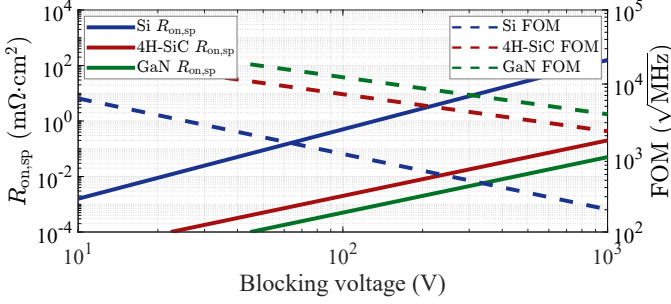


Fig. 2. Ideal one-dimensional unipolar specific on-state resistance $R_{on,sp}$ (left axis, solid) and the resulting high-frequency figure of merit $FOM = 1/\sqrt{R_{on,sp}C_{oss,sp}}$ (right axis, dashed, in $\sqrt{\text{MHz}}$ – the product $R_{on,sp}C_{oss,sp}$ is an area-independent RC time constant) versus blocking voltage for Si, 4H-SiC, and GaN.

B. Semiconductor Blocking-Voltage Tradeoff

In a conventional two-level voltage-source inverter, each active switch must block a voltage directly tied to the complete dc-link voltage. In unipolar power devices, a higher blocking-voltage rating imposes a fundamental tradeoff against both specific on-state resistance and switching performance, commonly captured by the figure of merit, as shown in Fig. 2 [21], [22]. Therefore, a lower-voltage device consistently achieves a higher FOM than a higher-voltage device of the same material.

Wide-bandgap (WBG) semiconductors shift this FOM favorably: SiC metal-oxide-semiconductor field-effect transistors (MOSFETs) now dominate high-voltage automotive drives, while GaN is particularly attractive at lower voltage owing to its fast switching behavior and superior FOM [21], [23]–[25]. Yet switching material does not remove the underlying coupling between dc-link voltage and device voltage class: irrespective of technology, a higher dc-link voltage still forces a higher-voltage, higher- $R_{on,sp}$ device and higher switching losses. Converter architectures that partition the dc-link voltage among multiple switching units, so each device blocks only a fraction of the total voltage, therefore become increasingly relevant as system voltage rises.

C. Multilevel Converters and Their Limitations

Multilevel converters constitute an established solution to this voltage-partitioning problem. NPC, T-type, flying-capacitor, and cascaded architectures all distribute the dc-link voltage among additional semiconductor positions or capacitor-voltage levels, reducing individual device voltage stress [26]–[29].

This voltage partitioning, however, comes at the cost of additional semiconductor devices, passive components, and gate drivers – each an additional potential failure point – without, by itself, yielding a modular converter–machine system: because the added semiconductor positions continue to feed a single, electrically common three-phase winding set, a fault at one position cannot generally be isolated and bypassed without disabling the drive. Fault tolerance, power redistribution, and post-fault reconfiguration can be implemented in conventional multilevel drives, but typically require additional redundancy, dedicated isolation paths, and specialized control layered on top of the base topology rather than following from it. A larger semiconductor count therefore does not yield a commensurate increase in fault tolerance, motivating an alternative form of voltage partitioning.

D. Stacked Polyphase Bridge

Unlike other multilevel converters, which distribute the dc-link voltage among additional semiconductor positions within a single converter, the stacked polyphase bridge (SPB) distributes the high-voltage conversion function among N repeated converter units. Each unit is a conventional three-phase two-level bridge connected in series on the dc side, supplying its own electrically separated three-phase winding group of the same machine, as illustrated in Fig. 3.

Under balanced operation, each cell’s local dc-link voltage $V_{dc,k} \approx V_{dc}/N$, so the semiconductor voltage rating can be selected on the local cell voltage rather than the complete traction dc-link voltage – addressing the semiconductor bottleneck much as multilevel converters do. Thanks to each cell also drives its own separated winding group, SPB additionally has an inherently modular structure that multilevel converters lack. A faulted cell can, in principle, be isolated and bypassed while the remaining cells continue to drive the machine at reduced power. It is this combination of semiconductor voltage reduction and converter–machine modularity that distinguishes SPB from conventional multilevel and cascaded architectures and motivates its consideration as a promising candidate topology for next-generation traction drives.

Realizing this potential, however, introduces coupled design challenges in component count, dc-link dynamics, machine design, common-mode voltage, and fault handling, addressed throughout this review. These challenges have grown more tractable as neighboring research areas have matured: multiphase machines provide additional post-fault degrees of freedom, integrated modular motor drives show how repeated power-electronic units can be physically associated with indi-

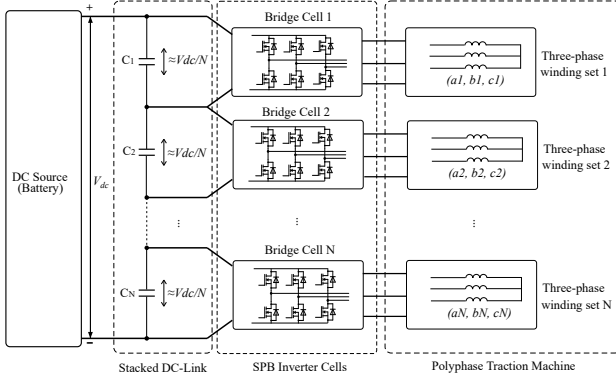


Fig. 3. Conceptual architecture of an N -cell SPB traction drive: series-connected local dc-link capacitors, series-connected three-phase two-level bridge cells, and their electrically separated machine winding groups.

vidual machine sections, wide-bandgap semiconductors amplify performance differences among voltage classes, and distributed control coordinates modular converter units [30]–[35].

Despite more than a decade of SPB research, beginning with Norrga *et al.* [36], no study has yet brought these threads together to systematically assess its feasibility for electric-vehicle traction applications. The available literature remains distributed across converter topology, modulation, dc-link stability, multiphase-machine design, integrated motor drives, semiconductor technology, thermal management, distributed control, and fault-tolerant operation. Although these topics are strongly coupled in an SPB traction drive, they are rarely evaluated within a common system boundary, leaving a gap between a well-motivated architectural concept and a system-level feasibility assessment. Accordingly, this paper provides a system-level review of SPB traction-drive technology aimed at closing that gap.

The remainder of the paper is organized as follows. First, SPB’s historical development, operating principle, and modulation are consolidated. Second, the interaction between cell count and the electric machine is examined alongside multiphase-machine and integrated-modular-drive research, including the insulation-coordination overhead of series-connected cells. Third, common-mode voltage and dc-link dynamics and balancing are examined. Fourth, semiconductor voltage scaling and fault handling are treated as coupled architecture-level design problems. Finally, the remaining research gaps and automotive validation requirements are identified.

II. EVOLUTION OF THE STACKED POLYPHASE BRIDGE CONCEPT

SPB belongs to the broader development of modular converter–machine systems. Modular and integrated modular motor drives established repeated converter–winding units, machine segmentation, power sharing, and fault-tolerant operation [30], [37]. Series-connected implementations extended these concepts to dc-voltage sharing, enabling lower-voltage semiconductor devices [31], [38], [39]. Other modular drive topologies adopted the same principle [40], [41].

Norrga *et al.* introduced SPB in 2013 [36], followed by dedicated control and modulation methods [42]. Related series-connected drives demonstrated internal dc-link regulation using machine-side degrees of freedom [43]. Subsequent stability analyses established the balancing-controller requirements [44], [45], identifying dc-link balancing as a central control challenge alongside semiconductor voltage selection.

Research then expanded toward coordinated modulation, scalable control, and converter–machine design. Modulation studies linked switching loss to dc-link harmonics [46], while distributed control provided an alternative to centralized balancing [47]. Comparative assessment against parallel-connected polyphase bridges and other modular converters evaluated machine design, losses, capacitor energy, cost, and redundancy, positioning SPB among several candidate architectures [48]. Interleaving and phase-count studies further connected converter performance to machine architecture [33], [49]–[51].

Recent work has extended SPB toward decentralized and reconfigurable operation. Common-duty-ratio control, decentralized balancing, and multi-agent power distribution reduce reliance on centralized coordination [52]–[55]. Online module isolation and reintroduction enable dynamic dc-bus reconfiguration, including removal of a faulted cell without interrupting torque production [56]. In parallel, gallium nitride devices have renewed interest in the benefits of lower semiconductor voltage classes [57].

Fig. 4 groups these developments by research focus, tracing the progression from voltage partitioning to coordinated converter–machine design and reconfigurable operation. The remaining integration gap for electric vehicles is a unified, application-specific SPB design framework that couples battery and dc-link requirements, semiconductor selection, machine configuration, and control, evaluated against common energy, power-density, cost, and reliability targets.

III. OPERATING PRINCIPLE AND TOPOLOGICAL CHARACTERISTICS

A. Basic Structure and Voltage Partitioning

An N -cell SPB consists of N conventional three-phase two-level voltage-source inverter cells connected in series on the dc side, with each bridge supplying one electrically separated three-phase winding group of the same machine [36], [42], [46]. The machine therefore presents N ac ports, while the converter cells share a single series dc path.

The dc-link voltage satisfies

$$V_{dc} = \sum_{k=1}^N V_{dc,k}, \quad V_{dc,k} \approx \frac{V_{dc}}{N} \quad (1)$$

under balanced operation. A basic implementation contains $6N$ active switches and N local dc-link capacitors. Increasing N reduces the nominal cell voltage, and hence the required semiconductor voltage class, but simultaneously increases the number of switches, gate drivers, sensors, capacitors, and machine-winding terminals that must be realized and managed. Fig. 5 shows this structure at switch level for two representative cells, making explicit the $6N$ -switch bridge realization,

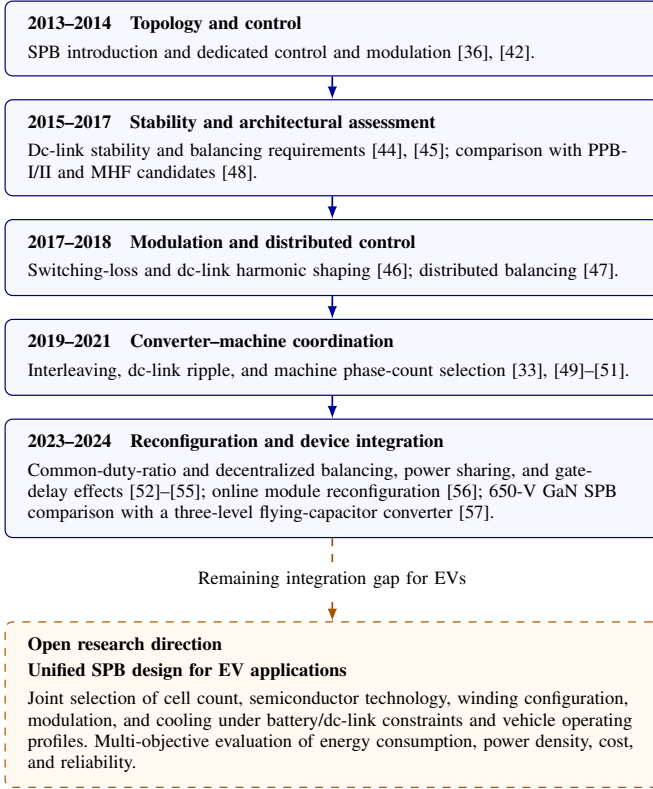


Fig. 4. Representative developments in SPB research grouped by period and research focus. Overlapping periods reflect concurrent progress in stability, modulation, control, and converter–machine integration. The dashed block identifies the EV-specific research direction synthesized in this review.

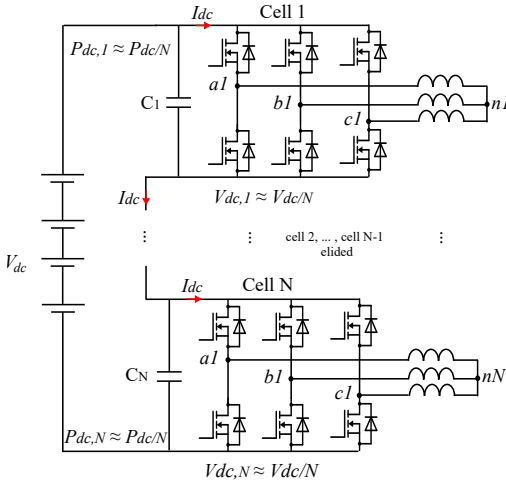


Fig. 5. Switch-level realization of the SPB dc–ac stage for two representative cells (Cell 1 and Cell N ; cells $2, \dots, N-1$ elided). The common series current i_{dc} (2) flows through every cell, giving each cell the dc-side power $p_{dc,k} = V_{dc,k} i_{dc}$ (3).

the local capacitors C_k , and the notation used throughout the remainder of this paper.

B. Series DC Connection and Power Flow

Neglecting auxiliary and bypass paths, the cell dc currents are constrained to be equal,

$$i_{dc,1} = i_{dc,2} = \dots = i_{dc,N} = i_{dc}, \quad (2)$$

so that the dc-side power entering cell k is

$$p_{dc,k} = V_{dc,k} i_{dc}. \quad (3)$$

The total machine power, by contrast, is the sum of the powers delivered to the individual winding groups,

$$P_{ac} = \sum_{k=1}^N P_k. \quad (4)$$

Under symmetrical operation $P_k \approx P_{ac}/N$, but equal power sharing is not a fundamental requirement of the topology. The multi-three-phase machine provides additional current and power degrees of freedom that allow power to be redistributed among winding groups [43], [54]. This redistribution capability is the physical basis for both dc-link voltage balancing and several of the post-fault operating strategies.

The combination of a common dc current with independently controllable ac-side powers also explains why the local capacitor voltages must be treated as internal dynamic states rather than fixed design parameters. If one winding group processes more power than another, the corresponding capacitor energies can diverge even though all cells carry the same dc current [44], [45]. Managing this divergence is the dc-link voltage-balancing problem, SPB's first-order control challenge.

C. Cell Count as a Converter–Machine Design Variable

Cell count is consequently a converter–machine design variable rather than a semiconductor variable alone. Increasing N also reduces the voltage available to each winding group and alters the number of winding sets, terminal count, electromagnetic coupling, current-control degrees of freedom, and available reconfiguration states.

Machine phase current, however, does not generally scale as $1/N$: it instead depends on the number of turns, flux linkage, winding connection, torque constant, and machine geometry. A many-cell architecture may therefore reduce the semiconductor voltage class while still requiring high-current devices or parallel chips at the device level – a coupling among cell count, voltage class, and current rating that matters most as the design space extends toward lower-voltage device classes.

IV. ELECTRIC-MACHINE REQUIREMENTS AND INTEGRATED IMPLEMENTATION

A. Winding Architecture and Electromagnetic Coupling

Although the winding groups are electrically separated at their terminals, they share the same magnetic circuit, rotor, air gap, and shaft. Their electromagnetic coupling depends on winding distribution, slot/pole combination, spatial displacement, and machine geometry [33], [43], [54], [58]. Therefore, an SPB-compatible machine cannot generally be obtained simply by dividing a conventional three-phase winding into multiple groups without considering the resulting coupling and harmonic behavior.

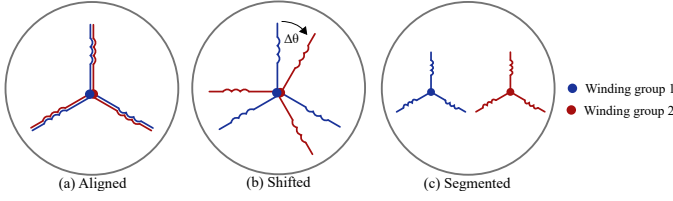


Fig. 6. Schematic comparison of winding-group spatial arrangements, each coil marking one phase axis of winding group 1 (blue) or group 2 (red): (a) aligned groups (parallel phase axes), (b) spatially shifted groups (offset by $\Delta\theta$), and (c) physically segmented stator modules.

The three-phase groups may be aligned, spatially shifted, or implemented as physically segmented stator modules. Spatial displacement affects mutual coupling, torque-ripple components, and the additional current subspaces available to the controller, among other effects [59]–[61]. These additional subspaces can introduce non-torque-producing currents, but they can also be exploited for harmonic suppression, differential power sharing, and fault-tolerant operation [54], [62]. Split winding architectures show a similar tradeoff: they can reduce ripple and provide additional current/voltage degrees of freedom, but they do so by increasing winding terminals and coupling complexity [63]. These three winding-group arrangements are compared schematically in Fig. 6.

For the k th PMSM winding group,

$$\begin{aligned} v_{d,k} &= R_{s,k} i_{d,k} - \omega_e L_{q,k} i_{q,k}, \\ v_{q,k} &= R_{s,k} i_{q,k} + \omega_e (L_{d,k} i_{d,k} + \psi_{f,k}), \end{aligned} \quad (5)$$

and the required voltage must satisfy

$$\sqrt{v_{d,k}^2 + v_{q,k}^2} \leq k_{\text{mod}} V_{\text{dc},k} \approx k_{\text{mod}} \frac{V_{\text{dc}}}{N}. \quad (6)$$

This falling per-cell voltage budget ties winding turns, inductance, and field-weakening capability directly to N – a coupling demonstrated in optimized multiphase SPB-PMSM systems [33]. Recent multiphase traction-machine studies similarly emphasize coordinated design of winding voltage, harmonic subspaces, and modulation [64], [65].

B. From Multiphase Machine to Integrated Modular Drive

Physical integration can convert the added electrical complexity of many winding groups into a packaging advantage. With a remote inverter, additional winding groups increase phase-cable and connector count; positioning each converter bridge close to its corresponding winding group instead reduces external cabling, shortens high-frequency interconnections, distributes heat sources, and aligns converter modules with stator segments.

This design philosophy predates SPB and has been extensively studied in integrated and modular motor drives [30], [31], [34], [38], [39], [66]–[78], summarized in Table II. The overall trend runs from early cable-reduction and compact-packaging concepts through jointly designed fault-tolerant converter/winding/thermal architectures and series-connected GaN IMMDs to recent WBG-based axial-flux IMMDs demonstrating stator-mounted converter modules and scalability to MW-class and EV-traction implementations.

TABLE II
REPRESENTATIVE INTEGRATED MOTOR-DRIVE DEVELOPMENTS

Reference	Integration theme
Shakweh <i>et al.</i> [66]	Integrated motor/inverter architecture
Wheeler/Pickering <i>et al.</i> [67], [68]	Converter and thermal design inside motor envelope
Brown <i>et al.</i> [69]	Converter mounted at modular stator pole
Wolmarans <i>et al.</i> [70]	Fault-tolerant converter, machine, and thermal design
Choi <i>et al.</i> [30]	Modular PM machine and high-temperature electronics
Wang <i>et al.</i> [31], [38]	Low-voltage GaN, dc-voltage sharing, and interleaving
Uğur and Keysan [72], [73]	Multi-physics converter-machine optimization
Mohamed <i>et al.</i> [74]–[76]	Stator-mounted modules, polygon housing, integrated dc link
Wang <i>et al.</i> [77]	Repeated machine and converter submodules
Chen <i>et al.</i> [78]	WBG converter, machine, passive, and cooling co-design
Deng <i>et al.</i> [34]	Recent IMD technologies and research directions

C. Physical Integration Options

The electrically separated winding groups required by SPB do not impose a unique converter placement on a radial-flux machine. Chowdhury *et al.* [79], consistent with the broader integrated-motor-drive survey of Abebe *et al.* [80], classify motor/inverter physical integration into four categories: whether the converter mounts radially or axially, and whether it interfaces with a separate housing/end-cap structure or directly with the stator core. This taxonomy frames how SPB converter cells can be placed around a radial-flux machine, as illustrated in Fig. 7. The axial-flux stator-mounted and polygon-housing alternative is covered by the Mohamed *et al.* developments [74]–[76] in Table II.

Modern WBG drives therefore provide several machine and packaging concepts that are compatible with SPB, including segmented windings, distributed converter cells, shared cooling, and integrated dc-link structures. These developments do not establish an intrinsic SPB advantage, but they demonstrate the availability of enabling technologies required for practical implementation.

D. Integration Is Not Free of Complexity

Integration does not remove complexity; it relocates it. Converter modules mounted on or near the machine are exposed to elevated temperature, vibration, and sealing/coolant requirements, among other stresses, and to electromagnetic coupling between gate-drive, power, and machine conductors. Shared machine-converter cooling can also increase thermal interaction between the stator and power electronics.

SPB introduces an additional insulation challenge because its converter cells and associated winding groups occupy different absolute potentials along the series dc stack. Gate-driver supplies, sensors, and mechanical mounting, among other subsystems, must therefore preserve the required electrical isolation.

Consequently, practical SPB integration requires coordinated electrical, thermal, and mechanical design, complementing the semiconductor- and system-level tradeoffs.

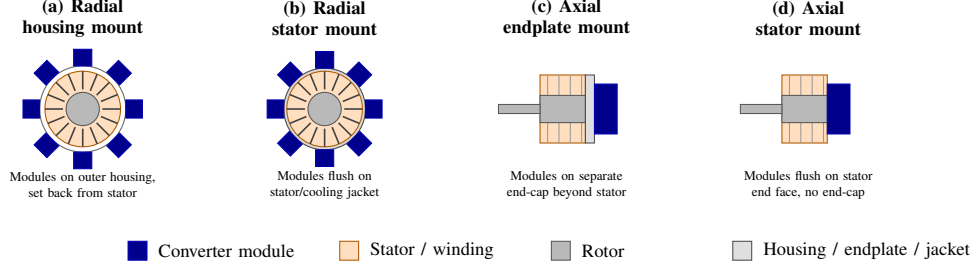


Fig. 7. Physical integration options for SPB-compatible converter modules on a radial-flux machine, following the four-category taxonomy of [79]: (a) radial housing mount, (b) radial stator mount, (c) axial endplate mount, and (d) axial stator mount, each shown in whichever radial or axial cross-section most clearly conveys its mounting geometry.

TABLE III
SPB MODULATION DEGREES OF FREEDOM GROUPED BY PRINCIPAL SYSTEM-LEVEL EFFECT.

Effect	Degrees of freedom
Ripple and loss shaping	Carrier phase [46], [51], [81]; switching sequence [82], [83]; clamping and DPWM [84], [85]
CMV and balancing authority	Zero-sequence injection and vector redundancy [86], [87]; differential cell commands [54], [88]

V. MODULATION STRATEGIES

Each stacked polyphase bridge (SPB) cell supports conventional sinusoidal pulsewidth modulation (SPWM), space-vector modulation (SVM), and discontinuous pulsewidth modulation (DPWM). However, the shared dc current, dynamic cell voltages, and electromagnetic coupling between winding groups require coordination across cells. Carrier phase, zero-sequence injection, switching sequence, and clamping intervals provide additional degrees of freedom for shaping dc-link ripple, machine harmonics, switching loss, common-mode voltage (CMV), and winding-group power distribution [42], [46]. Table III groups these degrees of freedom by their principal system-level effects.

A. Carrier Coordination and Harmonic Cancellation

The switching function of phase leg x in cell k has the double-Fourier expansion

$$\begin{aligned}
 s_{x,k}(t) = & \frac{A_{00}}{2} + \sum_{n=1}^{\infty} [A_{0n} \cos(n[\omega_{0,k}t + \theta_{0,k}]) + B_{0n} \sin(\cdot)] \\
 & + \sum_{m=1}^{\infty} [A_{m0} \cos(m[\omega_{c,k}t + \theta_{c,k}]) + B_{m0} \sin(\cdot)] \\
 & + \sum_{m=1}^{\infty} \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \left[A_{mn} \cos(m[\omega_{c,k}t + \theta_{c,k}] \right. \\
 & \quad \left. + n[\omega_{0,k}t + \theta_{0,k}]) + B_{mn} \sin(\cdot) \right], \quad (7)
 \end{aligned}$$

where m and n denote the carrier and sideband indices, respectively; $\omega_{c,k}$ and $\theta_{c,k}$ are the carrier frequency and phase; and $\omega_{0,k}$ and $\theta_{0,k}$ are the fundamental frequency and phase. Each sine term has the same argument as its corresponding cosine term.

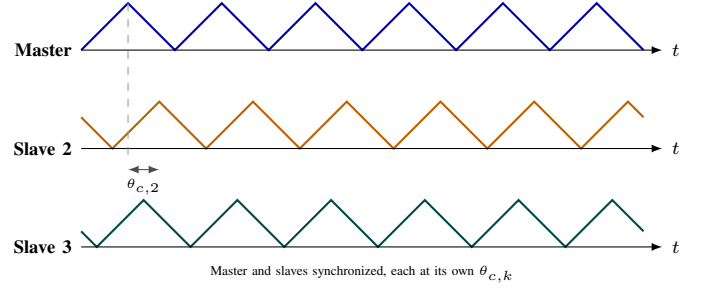


Fig. 8. Master/slave carrier synchronization for $N = 3$: each slave carrier has a prescribed phase offset $\theta_{c,k}$ relative to the master, according to (8).

The carrier phase is a free parameter for shaping aggregate harmonics. Uniform interleaving assigns

$$\theta_{c,k} = \theta_{c,1} + (k-1)\Delta\theta_c, \quad \Delta\theta_c = \frac{2\pi}{N}. \quad (8)$$

Fig. 8 illustrates this arrangement for three cells.

B. DC-Link Ripple and Battery-Side Current

DC-link current ripple is a design concern because of its influence on battery aging. Lower-voltage switches typically permit higher switching frequencies within a given loss budget, reducing per-cell capacitance requirements and battery-side ripple relative to a single high-voltage bridge. Carrier interleaving provides further ripple reduction.

The dc-side current spectrum is obtained by convolving each phase switching-function spectrum with its current spectrum,

$$I_{\text{conv},x,k} = \frac{1}{2} (S_{x,k} \otimes I_{x,k}). \quad (9)$$

and summing the three phase contributions. For a single cell, carrier harmonic m produces sidebands at $n = \pm 6p - 3$ for odd m and $n = \pm 6p$ for even m , where p is an integer. With uniform interleaving, $\theta_{c,k} = 2\pi k/N$, the summed converter currents retain carrier orders $m = pN$ for even N and $m = 2pN$ for odd N , with the corresponding sideband structure [42].

The shared dc-link current follows from the equivalent circuit developed in Section VII:

$$L \frac{di_{\text{dc}}}{dt} = E - Ri_{\text{dc}} - \sum_k v_{\text{conv},k}, \quad C_k \frac{dv_{\text{conv},k}}{dt} = i_{\text{dc}} - i_{\text{conv},k}, \quad (10)$$

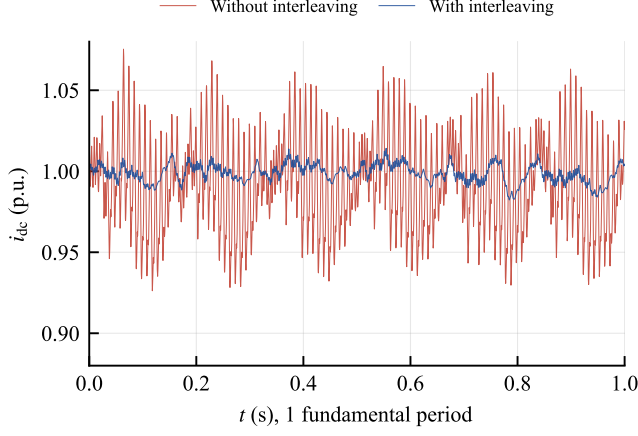


Fig. 9. Simulated shared dc-link current $i_{dc}(t)$ for three series-connected SPB cells, with $M = 0.85$, $f_1 = 1$ Hz, and $f_{sw} = 100$ Hz. Without interleaving (red) and with uniform carrier interleaving (blue), the currents have the same dc average; interleaving reduces peak-to-peak ripple by approximately 4–5 $\times$, consistent with (12). Results are from a coupled circuit simulation, not hardware measurements.

For equal cell capacitances, $C_k = C$, the relationship between the summed converter-current spectrum and dc-bus current is [42]

$$I_{dc}(\omega) = \frac{\sum_{k=1}^N I_{conv,k}(\omega)}{N - \omega^2 LC + j\omega CR}. \quad (11)$$

Approximating the residual ripple by the lowest surviving carrier harmonic, $m = N$, gives

$$\Delta i_{dc} \approx \frac{\left| \sum_{k=1}^N I_{conv,k}(2\pi N f_{sw}) \right|}{\left| N - (2\pi N f_{sw})^2 LC + j2\pi N f_{sw} CR \right|}. \quad (12)$$

Interleaving therefore combines cancellation of lower-order components with greater LC attenuation at the retained frequency, Nf_{sw} . Fig. 9 illustrates the resulting reduction for $N = 3$.

The simulation includes lumped battery-to-stack inductance but assumes ideal cell connections. In distributed implementations, bus-bar and connector-loop inductances alter harmonic cancellation. A series-connected gallium nitride integrated modular motor drive required careful capacitor selection and compact layout to preserve interleaving benefits [39]; an earlier characterization found that parasitic loop inductance could make interleaved ripple exceed the noninterleaved value [89]. The illustrated reduction should therefore be treated as an upper bound: achievable performance also depends on layout, capacitor selection, and switching frequency.

C. Machine Harmonics, Voltage Utilization, and Switching Loss

The preferred carrier displacement depends on cell count, modulation index, power factor, winding displacement, and the optimization objective. SPB studies demonstrate its influence on semiconductor losses and dc-link current harmonics [46], while related modular and multiphase drives demonstrate simultaneous reductions in dc-link and machine current ripple [51], [82], [90].

In multiphase machines, nonfundamental voltages can excite low-impedance harmonic subspaces, increasing copper loss and torque ripple despite satisfactory fundamental current control. Interleaved and discontinuous modulation strategies therefore require adaptation to the multiphase current-ripple characteristics [81], [91], [92].

At high speed, voltage utilization becomes critical because each winding group is supplied by its local cell voltage $V_{dc,k}$, subject to the machine voltage limit in (6). This constraint becomes more restrictive as cell count increases and local voltage decreases. Optimized vector selection in dual-three-phase traction machines can extend operation in the high-modulation region while suppressing unwanted subspace currents [64].

Switching-loss reduction introduces a competing objective. DPWM and hybrid strategies reduce transitions by clamping selected phases or redistributing high-frequency switching among modules. Related open-end-winding studies demonstrate simultaneous reductions in switching loss and zero-sequence current [84], [85]. Although their dc architecture differs, the principle of coordinating switching activity among inverter modules is applicable to SPB.

D. Coupling to Voltage Balancing and Common-Mode Control

Modulation also determines the control authority available for dc-link voltage balancing and CMV reduction. Voltage balancing requires differential winding-group power commands that remain feasible within local voltage and current limits [54], [55], [88]. Redundant voltage vectors and coordinated zero-sequence injection can reduce common-mode excitation of the machine frame [86], [87], [93]. These objectives are examined in Sections VII and VI, respectively.

SPB modulation consequently requires coordinated switching decisions that account for ripple, losses, voltage utilization, balancing, and common-mode behavior across the operating map.

VI. COMMON-MODE VOLTAGE EFFECTS

In any converter-fed ac machine, the non-zero average of the three phase-leg pole voltages produced by PWM switching gives rise to a common-mode voltage (CMV) that excites parasitic capacitive paths between winding, frame, shaft, and bearings. This mechanism is well characterized for a converter with a single machine port whose voltage scaling is produced internally, such as the flying-capacitor multilevel converter, for which an analytical CM-EMI model referenced to one dc-link potential exists [94]. SPB does not fit this picture: because its N cells are connected in series on the dc side (Section III) while each feeds an electrically separate winding group, cell k 's local dc rails are displaced from the overall dc-bus reference by an amount that depends on its position in the stack.

A. Series-Stack Origin of Common-Mode Excitation

Taking cell 1's negative dc rail as the reference, with cell k 's negative rail coinciding with cell $(k - 1)$'s positive rail, the absolute offset of cell k is

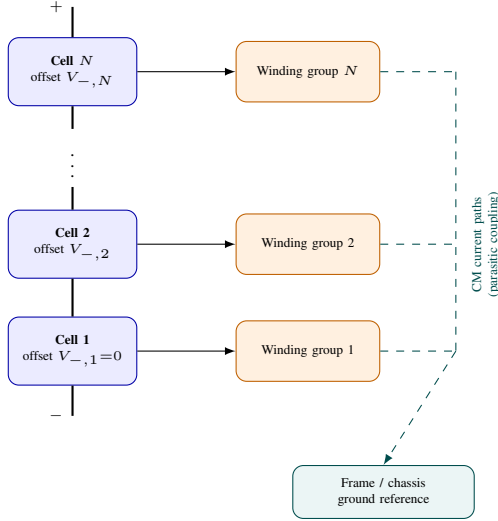


Fig. 10. Simplified SPB series-stack structure: absolute potential offset per cell (13) and representative CM current paths (dashed) to the shared frame/chassis ground through parasitic capacitance.

$$V_{-,k} = \sum_{j=1}^{k-1} V_{dc,j} \approx (k-1) \frac{V_{dc}}{N}, \quad (13)$$

using the voltage-partitioning relation (1). The common-mode potential seen by winding group k is then

$$v_{cm,k}^{abs}(t) = V_{-,k} + \frac{1}{3}(v_{aN_k} + v_{bN_k} + v_{cN_k}), \quad (14)$$

where $v_{aN_k}, v_{bN_k}, v_{cN_k}$ are cell k 's pole voltages referenced to its own negative rail N_k . The second term is the conventional switching-induced CMV already addressed by zero-sequence injection and vector-redundancy modulation (Section V); the first term, $V_{-,k}$, is specific to SPB, set by the cell's stack position and by the instantaneous voltage-sharing state (Section VII), not by modulation alone.

Fig. 10 illustrates the resulting structure: increasing absolute potential offset with stack position, together with representative CM current paths coupling each winding group to the shared frame and chassis ground through parasitic capacitance. Because the cells and their winding groups are physically close together in the integrated implementations of Section IV, these paths are short and the coupling between cells is direct rather than mediated by long cabling. Physically integrated motor-drive assemblies show that common-mode current can resonate through exactly this kind of parasitic loop, and that the resonance is not adequately predicted by a single-source CM model [93], [95] – extending such modeling to a multi-port structure with N distinct dc offsets is, to the authors' knowledge, not yet available for SPB specifically.

B. Carrier Synchronization for CMV Cancellation

Beyond the stack-offset mechanism above, synchronizing each cell's PWM waveform relative to the others is a way to actively cancel the switching-induced component of CMV. We illustrate the principle for the $N = 2$ case, where the

cancellation condition has a simple closed form; the same synchronization idea extends to larger stacks built from matched cell pairs, though a general N -cell synchronization law has not yet been derived for SPB.

If cell 2 is driven with the logical inverse of cell 1's gate commands, and cell 2's winding set is wired with reversed winding sense so that both cells still generate torque in the same direction, the two cells' individually switched CMVs become approximately equal and opposite, $v_{cm,1}(t) \approx -v_{cm,2}(t)$, and the total CMV seen by the machine,

$$v_{cm,tot}(t) = \frac{v_{cm,1}(t) + v_{cm,2}(t)}{2}, \quad (15)$$

cancels toward zero instead of adding constructively. This balanced-cancellation principle was first demonstrated for a dual-winding-set machine driven by a balanced inverter pair on a single, non-series-stacked machine [96], and was later applied to a genuine two-cell series-stacked (2L-SSC) SPB inverter [57]; as with interleaved GaN IMMDs, the achievable reduction is ultimately limited by layout parasitics and local dc-link capacitor design [39].

This cancellation assumes perfectly matched, simultaneous switching transitions between the two cells: a propagation-delay mismatch (e.g., from PCB layout or driver differences) reintroduces a residual CM spike, of duration equal to the delay, at every misaligned transition, and unequal power sharing during dc-link balancing similarly degrades the cancellation, both shown quantitatively in [57], [96]. The mechanism is specific to $N = 2$, or to a stack of matched cell pairs, so it should be read as a valuable special-case mitigation rather than a general solution to the stack-offset problem above. Fig. 11 simulates this cancellation directly from switching-function waveforms, including its effect on the resulting bearing/ground current.

C. Physical Consequences and Mitigation

CMV-driven bearing current is a documented failure mechanism in inverter-fed machines generally, arising from capacitive coupling through the air gap and discharge events once the bearing lubricant film breaks down [97], and can be identified with dedicated non-intrusive current-measurement techniques [98]. In SPB, this mechanism acts at each of the N winding groups individually; because (13)–(14) gives each group a different absolute CM reference, the resulting bearing- and shaft-voltage stress is not necessarily uniform across the stack, and whether it is dominated by the highest-offset cell, by all cells combined, or by resonant interaction between cells remains open. Shaft-voltage cancellation circuits developed for conventional drives [99] are a plausible mitigation candidate but have not been demonstrated for a series-stacked, multi-port machine.

The same switching events also produce fast dv/dt transitions at the winding terminals, stressing turn-to-turn and phase-to-frame insulation independently of the CM current path. WBG devices, already motivated for SPB by their favorable scaling at reduced blocking voltage (Section VIII), shorten rise time and increase overshoot, and have been shown to accelerate insulation aging and partial-discharge activity

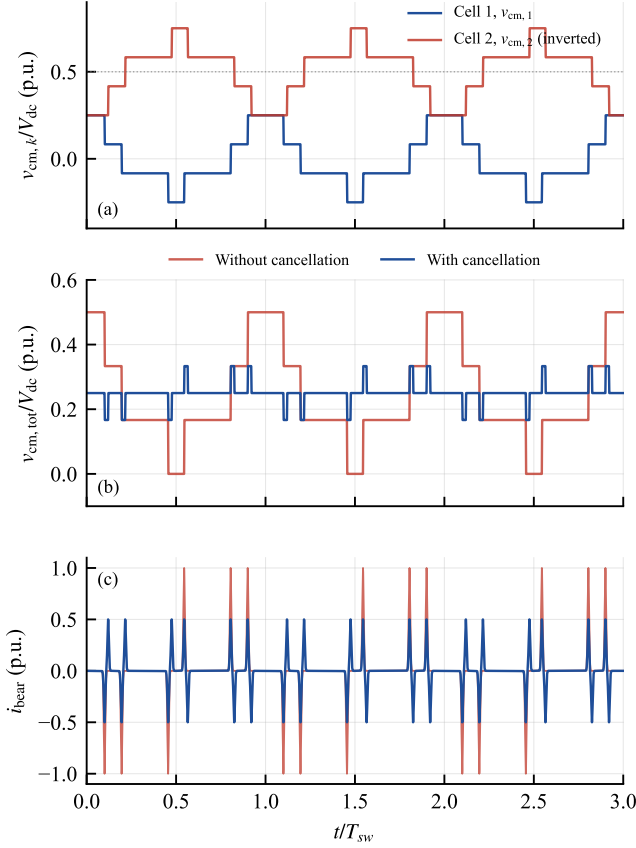


Fig. 11. Switching-function simulation of $N = 2$ SPB common-mode cancellation ($M = 0.85$, $\cos \varphi \approx 0.94$), all quantities in per unit, with cell 2 driven by the logically inverted gate commands of cell 1 plus a propagation delay $t_d = 0.02 T_{sw}$. (a) Each cell's absolute CM voltage $v_{cm,k}^{abs}$, combining the stack offset $V_{-,k}$ (13) with its local switching term. (b) Total CM voltage $v_{cm,tot}(t)$ (15): cancellation removes the switching ripple, leaving only delay-induced residual spikes on the same stack-offset average. (c) Resulting bearing/ground current $i_{bear} = C dv_{cm,tot}/dt$, normalized to the without-cancellation peak. Computed simulation result, not measured hardware data.

in hairpin and other traction-machine windings [100]–[102]; diagnostic methods exist to assess the resulting insulation condition [103], within the broader set of insulation technologies for transport-electrification machines [104]. In SPB, this stress is superimposed on the stack-position-dependent offset of (13), so insulation coordination must account for both the local switching transition and the cell's absolute stack position, consistent with the challenge identified qualitatively in Section IV.

Mitigation therefore has to act at more than one level. Active EMI filtering can attenuate common-mode current at the converter terminals without relying on machine or bearing hardware [105], and cancellation circuits can target shaft voltage directly [99]. At the modulation level, Section V showed that zero-sequence injection and redundant-vector selection reduce CMV excitation [86], [87], but the carrier-phase displacement used there to interleave switching harmonics for ripple reduction, (8), is chosen to cancel a selected ripple harmonic and is not simultaneously optimized for common-mode excitation. Because SPB's CM problem additionally

depends on the fixed stack offset $V_{-,k}$, which carrier phasing cannot change, an interleaving choice that minimizes dc-link or stator ripple need not minimize aggregate common-mode excitation, and vice versa – a genuine SPB-specific design tension that future studies should address by extending the multi-objective modulation problem of Section V to common-mode and insulation-stress metrics evaluated across the full stack.

VII. DC-LINK DYNAMICS, STABILITY, AND VOLTAGE BALANCING

The common dc current and independently controlled winding-group powers (Section III) couple the local capacitor energies, so equal cell voltages are not inherently maintained [44], [45]. Related series-connected motor-drive research likewise shows that distributed current control alone does not ensure stable voltage sharing [89].

For cell k , the power balance is

$$C_k V_{dc,k} \frac{dV_{dc,k}}{dt} = V_{dc,k} i_{dc} - p_{ac,k} - p_{loss,k}. \quad (16)$$

Because the series current is common, dc input power depends on local voltage, whereas the winding-group controller determines ac output power. Defining the deviation from the mean cell voltage as

$$\tilde{V}_k = V_{dc,k} - \frac{1}{N} \sum_{j=1}^N V_{dc,j}. \quad (17)$$

a simplified differential-mode model around a balanced operating point is

$$C V_0 \frac{d\tilde{V}_k}{dt} \approx I_{dc,0} \tilde{V}_k - \Delta P_k. \quad (18)$$

where C and V_0 are the common cell capacitance and equilibrium voltage, $I_{dc,0}$ is the equilibrium dc current, and ΔP_k is the differential output-power perturbation. During motoring, $I_{dc,0} > 0$, a positive voltage perturbation increases cell input power and reinforces the deviation if output power remains unchanged. During regeneration, this feedback reverses. This interpretation agrees with SPB stability analyses [44], [45] and related series-connected multiphase-drive studies [106].

The constant-power-load model considered in [45] comprises

$$\begin{aligned} L_b \frac{di_{dc}}{dt} &= E_b - R_b i_{dc} - \sum_k V_{dc,k}, \\ C_k \frac{dV_{dc,k}}{dt} &= i_{dc} - \frac{P_k}{V_{dc,k}}. \end{aligned} \quad (19)$$

Here, E_b is the dc-source voltage, and L_b and R_b are the lumped series inductance and resistance of the shared source-to-stack supply path, including source and interconnection contributions; P_k is the cell's constant power demand. Under this model's assumptions, linearization about the open-loop motoring equilibrium for $N > 1$ yields $N - 1$ strictly positive real eigenvalues, establishing instability irrespective of parameter selection.

Fig. 12 illustrates the nonlinear dynamics using $L_b = 8 \mu\text{H}$, $R_b = 0.35 \Omega$, and $C = 100 \mu\text{F}$ per cell, taken from [45].

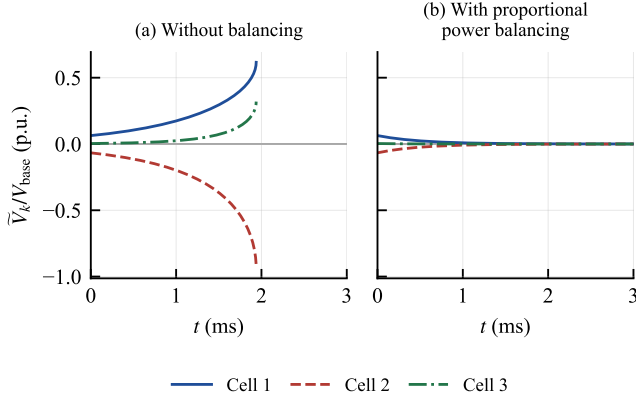


Fig. 12. Simulated normalized deviation from the instantaneous mean cell voltage, $\tilde{V}_k/V_{\text{base}}$, for $N = 3$: (a) without balancing and (b) with proportional power balancing motivated by [45]. Both cases use $E_b = 300$ V, $V_{\text{base}} = 100$ V, nominal cell power 1 kW, $V_{\text{dc},k}(0) = \{108, 95, 102\}$ V, and $i_{\text{dc}}(0) = 10$ A; $K_p = 30$ W/V in (b). Supply-path and capacitance parameters are given in the text. The open-loop trace ends when a cell reaches the 1-V termination threshold. Results are computed, not measured hardware data.

Without balancing, the initial mismatch grows until one cell approaches voltage collapse. A proportional differential-power law, $\Delta P_k = K_p \tilde{V}_k$, motivated by controller alternative I in [45], suppresses the imbalance on a comparable timescale. This simulation assumes ideal power-command tracking; it does not reproduce the complete current-control implementation.

The winding groups provide control authority over internal capacitor energy through differential power commands:

$$P_k^* = \frac{P^*}{N} + \Delta P_k, \quad \sum_{k=1}^N \Delta P_k = 0, \quad (20)$$

This allocation redistributes cell power without changing the total machine-power reference. Available balancing authority depends on phase-current margin, winding voltage, torque-ripple limits, and thermal state.

Table IV summarizes three principal approaches. Centralized control explicitly regulates voltage sharing through differential commands [44], [45]. Common-duty-ratio (CDR) control instead exploits the coupled converter-machine dynamics; stable sharing has been demonstrated for the investigated induction-machine system [52], with practical performance sensitive to timing mismatch [55]. Distributed multi-agent control extends local coordination to voltage balancing, power sharing, and reconfiguration [53], [54], [56]. Related modular-drive work uses local speed-droop laws to coordinate module power sharing [107].

For automotive applications, balancing must remain effective during torque transients, field weakening, parameter mismatch, and module reconfiguration. Sensor offsets, communication latency, propagation delays, and differential power limits determine the usable cell-voltage margin and therefore influence semiconductor voltage-rating selection. Dc-link stability must consequently be assessed across the coupled converter, machine, and control system.

TABLE IV
PRINCIPAL SPB DC-LINK BALANCING APPROACHES AND THEIR MAIN CONSIDERATIONS.

Approach	Mechanism	Key consideration
Centralized	Differential current/power commands [44], [45]	Direct authority; global sensing and coordination
CDR	Common modulation, natural coupled dynamics [52], [55]	Low overhead; symmetry- and timing-sensitive
Distributed	Local agents, neighbor communication [53], [54], [56]	Scalable; requires communication and synchronization

VIII. SEMICONDUCTOR TECHNOLOGY AND CELL-COUNT SCALING

SPB allows the use of lower-voltage semiconductor classes as N increases and the device rating must be selected from the maximum expected blocking stress, not the nominal voltage partition alone:

$$V_{\text{rated}} \geq k_{\text{rel}} \left(\frac{V_{\text{dc,max}}}{N} + \Delta V_{\text{imb}}^{\text{max}} + \Delta V_{\text{sw}}^{\text{max}} \right), \quad (21)$$

where $\Delta V_{\text{imb}}^{\text{max}}$ denotes the maximum cell-voltage imbalance, $\Delta V_{\text{sw}}^{\text{max}}$ the commutation overshoot, and k_{rel} the selected reliability margin. Cell count is therefore linked to discrete commercial voltage classes rather than to a continuous semiconductor-scaling law.

As previously discussed, SiC currently provides the most direct route for high-voltage traction, whereas GaN becomes increasingly attractive once the architecture reduces the per-device voltage enough to access lower-voltage classes with favorable conduction and switching characteristics [24], [35], [57].

For 1200-V dc links which is projected for future traction drives, the resulting discrete design transitions are summarized and visualized directly in Fig. 13.

Such as case of $N = 2$ gives a nominal cell voltage of 600 V and therefore leaves limited transient margin for a 650-V class. Increasing the cell count to $N = 3$ reduces the nominal cell voltage to 400 V and provides a substantially more practical operating region for the 650-V class. Further increases in N give access to progressively lower-voltage device families, as labeled directly on Fig. 13. The potential semiconductor benefit is therefore nonlinear because a small change in N can open a new commercial voltage class, whereas the active-switch count grows approximately linearly, as plotted directly from these design points in Fig. 13.

A. Commercial-Device Screening

Access to a lower voltage class does not by itself imply a superior performance: the comparison must be based on commercial devices and their conduction losses, switching performance in addition to thermal, and packaging constraints. Simple screening metrics such as $R_{\text{DS(on)}}C_{\text{oss}}$ or $R_{\text{DS(on)}}Q_g$ are useful for identifying candidate devices.

The principal device quantities required for SPB cell-count screening, and their primary role in the comparison,

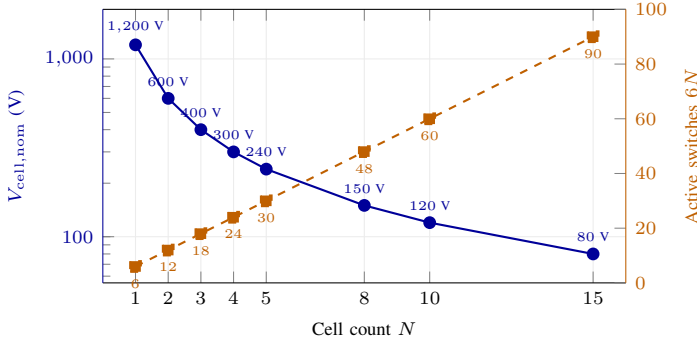


Fig. 13. Cell-count scaling for a 1200-V dc link: nominal cell voltage $V_{\text{cell,nom}} = V_{\text{dc}}/N$ (solid markers, left log axis) and active-switch count $6N$ (dashed markers, right linear axis) versus cell count N ; every marker is labeled directly with its exact value ($N = 1$: 1200 V, 6; $N = 2$: 600 V, 12; $N = 3$: 400 V, 18; $N = 4$: 300 V, 24; $N = 5$: 240 V, 30; $N = 8$: 150 V, 48; $N = 10$: 120 V, 60; $N = 15$: 80 V, 90). Increasing N opens progressively lower commercial semiconductor voltage classes (e.g., 1700–2000-V SiC at $N = 1$, 650-V GaN/SiC at $N = 3$, 100-V GaN/MOSFET class at $N = 15$) while the active-switch count grows linearly as $6N$.

are: $R_{\text{DS(on)}}(T)$, which sets the die-area or parallel-device requirement; $E_{\text{oss}}(V)$, which sets the output capacitance loss during switching; $E_{\text{on/off}}(V, I, T)$, which forms the overlapping switching-loss model; gate charge Q_g and Q_{gd} , which govern driver sizing and auxiliary loss; continuous current capability I_{cont} and thermal resistance R_{θ} , which govern parallelization and cooling requirements; and package or power loop inductance L_{σ} , which sets the usable blocking-voltage margin against di/dt -induced overshoot.

Current capability is particularly important because machine phase current does not generally decrease in proportion to $1/N$. If the derated current capability of one device is $I_{\text{dev,der}}$, the required number of parallel devices can be estimated as

$$n_p = \left\lceil \frac{I_{\text{sw,max}}}{I_{\text{dev,der}}} \right\rceil, \quad N_{\text{sw,tot}} = 6Nn_p. \quad (22)$$

Thus, a lower-voltage device may offer an attractive device-level FOM while still requiring substantial die area, package count, and gate-drive/thermal interfaces. Recent high-current GaN modules illustrate the importance of low-inductance packaging and integrated gate driving when GaN is scaled toward traction-current levels [108], [109].

Fast switching further couples device selection to layout. The commutation overshoot can be approximated by

$$\Delta V_{\text{sw}} \approx L_{\sigma} \frac{di}{dt}, \quad (23)$$

so the benefit of reduced nominal cell voltage can be lost if package and interconnect inductance consume a substantial fraction of the available blocking margin. This is particularly relevant for GaN, where high di/dt places strong requirements on the commutation loop, local decoupling, and gate-drive layout [109].

GaN device selection also requires consideration of dynamic $R_{\text{DS(on)}}$, reverse conduction, gate reliability, short-circuit capability, and thermal performance [110], [111]. These effects do not negate the benefit of lower blocking voltage, but they

prevent room-temperature static $R_{\text{DS(on)}}$ from being used as the sole criterion for cell-count selection.

B. Loss Scaling With Cell Count

For one device, the dominant semiconductor losses can be represented approximately as

$$P_{\text{cond}} \approx I_{\text{rms,sw}}^2 R_{\text{DS(on)}}(T_j), \quad (24)$$

and

$$P_{\text{sw}} \approx f_{\text{sw}} [E_{\text{on}}(V, I, T_j) + E_{\text{off}}(V, I, T_j)]. \quad (25)$$

At converter level,

$$P_{\text{semi}}(N) = \sum_{k=1}^N \sum_{j=1}^{6n_p(N)} (P_{\text{cond},k,j} + P_{\text{sw},k,j} + P_{\text{dt},k,j}) + P_{\text{gate}}(N). \quad (26)$$

The dependence on N is therefore not linear. The number of active devices increases with cell count, whereas $R_{\text{DS(on)}}$, switching energy, thermal impedance, and parallel-device requirement can change abruptly when the design enters a different semiconductor voltage class.

Consequently, SiC and GaN should be regarded as different regions of the SPB design space rather than as a fixed technology ranking. Low-cell-count architectures can favor high-voltage SiC because of its mature voltage and current capability, whereas increasing N can make lower-voltage GaN attractive if the improvement in conduction, switching, packaging, or passive-component performance compensates for the additional converter modules.

Resolving the cell-count question therefore requires reading Fig. 13 jointly with the screening quantities above and the machine requirements of, rather than from any single figure of merit.

IX. FAULT TOLERANCE AND RECONFIGURATION

SPB modularity provides structural redundancy because the converter and machine are divided into repeated converter-winding modules. However, fault tolerance is not automatic: because the dc ports are connected in series, a failed cell can disturb or interrupt the complete dc current path, a structural challenge demonstrated for SPB and for other series-connected modular converters alike [112]–[114]. Early SPB work demonstrated post-fault operation under semiconductor faults, establishing that repeated hardware must be accompanied by appropriate safe states and control action [112]. At the machine level, a submodule short-circuit fault in a modular converter feeding fractional-slot concentrated-winding PM machines was shown to drive a fault current above rated current because of the strong NdFeB rotor magnetization, risking machine overheating; reducing rotor magnetization limits this fault current but also narrows the constant-power-speed range, a design tradeoff specific to fault-tolerant FSCW PM machines [113]. In the SPB machine designs of that study, post-fault short-circuit current with NdFeB magnets exceeds the healthy-phase current, whereas with ferrite magnets it stays below it, at the cost of becoming visibly non-sinusoidal from magnetic coupling with the adjacent healthy windings – a

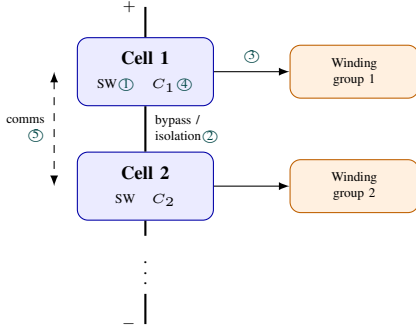


Fig. 14. Simplified SPB converter-machine structure with fault locations ①–⑤ annotated, corresponding in order to the rows of Table V.

concrete illustration that PM material selection is itself a fault-tolerance design variable, not only a torque-density one. Fault handling in other series-connected modular converters further underscores that mitigation requires the control system to be explicitly designed for module isolation [114].

The machine provides a second layer of redundancy. Healthy winding groups can potentially redistribute current after phase or converter faults, as demonstrated in recent multiphase and modular-drive studies [62], [115], [116]. In SPB, however, this machine-side redundancy is useful only if the failed converter cell can be isolated or bypassed without breaking the series dc path.

Building on the online reconfiguration capability introduced in Section II [56], after a faulted module is isolated the remaining active set $\mathcal{A}$ must satisfy

$$P^* = \sum_{k \in \mathcal{A}} P_k^*, \quad (27)$$

with torque derating if current, voltage, or thermal limits prevent full redistribution. Reconfiguration can therefore provide both fault isolation and an operating-range degree of freedom because the number of active cells changes the voltage available to the remaining winding groups. Isolating a cell also raises the voltage that each surviving cell must share of the fixed dc-link voltage V_{dc} ((1), now divided among $|\mathcal{A}| < N$ active cells), which is an additional reason current or power derating in (27) may be needed after a fault, as illustrated in Fig. 15. Fig. 14 maps these fault locations onto the SPB structure.

Reliability remains ambiguous because increasing cell count provides more potential degraded states but also more components that can fail. System or mission reliability is therefore more meaningful than component count alone. Markov-chain analysis has been applied to related fault-tolerant IMMDS to distinguish healthy, degraded, and failed states [118], while integrated distributed traction drives provide additional evidence that fault tolerance must include both the machine and the local converter architecture [70], [71].

The direct SPB literature therefore demonstrates the feasibility of post-fault operation and online reconfiguration, but not yet automotive-level fault coverage. Detection and isolation time, short-circuit energy, capacitor and sensor faults, communication loss, post-fault thermal loading, diagnostic

TABLE V
SPB FAULT LOCATIONS AND REQUIRED SYSTEM RESPONSE

Fault	SPB consequence	Required response
Switch / bridge leg	Cell cannot synthesize commanded voltage	Fast safe state and protection [112]
Converter module	Series dc path can be interrupted	Bypass/isolation and power redistribution [56]
Winding / phase	Reduced and asymmetric torque capability	Multiphase post-fault current control [62], [116]
Capacitor / sensor*	Loss of voltage-sharing information or margin	Detection, derating, isolation or shutdown
Communication / controller	Loss of distributed coordination	Local fallback and deterministic safe state [117]

*No SPB-specific study directly reports capacitor or sensor fault handling; this row reflects the authors' system-level inference from the general modular-converter requirement to detect loss of cell-voltage sensing or local dc-link capacitance and to derate or isolate the affected cell accordingly.

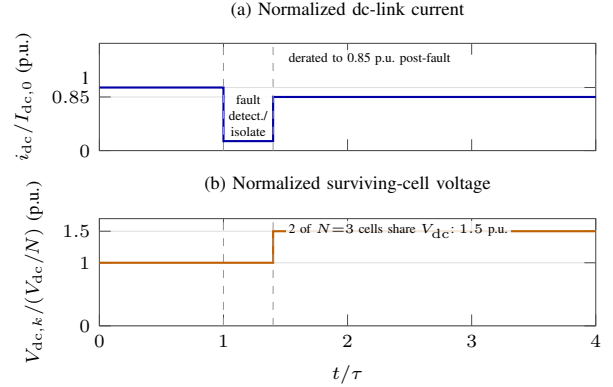


Fig. 15. Illustrative, analytically derived example of the post-fault current derating and remaining-cell voltage step-up for a representative $N = 3 \rightarrow 2$ case (one of three cells isolated after a fault); this is not measured or simulated data. (a) Normalized dc-link current, derated post-fault per (27). (b) Normalized remaining-cell voltage, stepping from 1 p.u. to $N/(N-1) = 1.5$ p.u. (1) once two cells must share V_{dc} . Both panels share the same time axis.

coverage, and deterministic functional-safety transitions remain important validation topics.

X. RESEARCH GAPS AND FUTURE DIRECTIONS

In theory, SPB is one of the more promising modular architectures for high-voltage electric traction: partitioning the dc-link voltage across series-connected cells grants access to lower-voltage semiconductor classes with a favorable figure of merit, supports modular converter-machine integration, and, as this review has shown, gives carrier-phase, balancing, and reconfiguration degrees of freedom that a conventional two- or three-level drive does not have. Whether these advantages translate into a real vehicle-level benefit, however, is not decided by any single subsystem but by how the battery/dc-link side, the semiconductor/converter side, and the machine side are co-designed together: the cell count that opens a favorable device voltage class also fixes winding voltage, current rating, and dc-link ripple frequency, and each of these choices feeds back into the others. This review has

assembled the pieces needed for that co-design – voltage partitioning and cell-count scaling (Sections III and VIII), machine-integration options (Section IV), modulation and dc-link ripple (Section V), common-mode behavior (Section VI), dc-link balancing stability (Section VII), and fault tolerance (Section IX) – but, like the literature it surveys, has largely treated them as separate design axes rather than jointly.

A. Toward an Integrated Co-Design Framework

Future work should combine the voltage-partitioning, modulation and ripple, dc-link balancing, common-mode, and fault-tolerating relations developed in this review into a single N - and technology-dependent framework, so that cell count, semiconductor class, and machine design can be evaluated jointly against vehicle-level energy, cost, and reliability targets rather than compared through separate, narrower studies. Several concrete open questions define what such a framework must resolve.

The optimum cell count N remains the central unresolved design question: as established in Section III, the voltage-versus-component-count tradeoff shifts the optimum as SiC, GaN, packaging, and cooling evolve [3], [24]. A common-platform comparison of a few-cell 650-V GaN SPB against a many-cell low-voltage GaN SPB, using real commercial devices, would be valuable; the GaN comparison in [57] is a useful starting point but should be extended to automotive power with current parallelization, machine voltage/speed constraints, dc-link capacitance, cooling, and mission energy.

Converter, machine, and thermal models, among others, should be co-optimized rather than designed sequentially. Existing phase-count studies show machine design changes with SPB cell number [33]; the missing step is a complete automotive co-design using vehicle-level energy, volume, mass, cost, and reliability as a joint objective.

Reliability is part of that objective: future studies should combine semiconductor, capacitor, sensor, gate-driver, communication, and winding faults (Section IX; [56], [112]) in one state-based, Markov-style framework as used for related fault-tolerant IMMDS [118], keeping *component failure*, *module isolation*, and *loss of vehicle function* distinct. An N -cell drive may survive with $N - 1$ active modules, but its torque–speed envelope, module thermal loading, and dc-voltage margin all change, so reliability data should pair with a post-fault capability model – not a series reliability block diagram – to compare availability, limp-home power, and mission completion probability against efficiency.

Cost and sustainability assessment should use the same N -dependent framing: increasing N can reduce per-switch die cost but tends to increase isolated gate supplies, sensing channels, and assembly operations, among others, almost linearly, while a repeated converter–winding module can gain from common tooling and automated assembly – a balance likely to differ between high-volume passenger vehicles and lower-volume heavy-duty systems. The same accounting should separate production, operation, maintenance, and end-of-life impact over one vehicle mission and service life, since modular repair and longer service life can offset the added production impact of extra components.

Overall, SPB research must move from demonstrating that the architecture operates toward demonstrating where it is better: whether it gives a measurable vehicle-level advantage over mature two- and three-level architectures once machine design, thermal management, EMC, reliability, manufacturing, and cost are included.

XI. CONCLUSION

This paper reviewed stacked polyphase bridge (SPB) converters – series-stacked two-level bridges each feeding an electrically separated winding group – as a candidate architecture for future high-voltage traction drives, partitioning the dc-link voltage while introducing converter–machine modularity, distributed control, and fault-management opportunities.

The review also shows that the semiconductor voltage advantage cannot be separated from the complete traction system. Increasing cell count adds converter, control, and insulation overhead, so SPB does not eliminate the complexity of high-voltage conversion but redistributes it between semiconductor technology, machine architecture, and control. WBG development makes this trade increasingly relevant, particularly where SPB can move a high-voltage traction system toward lower-voltage GaN devices; whether that approach is advantageous depends on whether the device-level gains compensate for the additional modular hardware and machine requirements.

The remaining step is not further isolated proof-of-concept studies but an integrated co-design across the battery/dc-link, semiconductor/converter, and machine sides, evaluated jointly against vehicle-level energy, cost, and reliability targets rather than compared through separate, narrower studies. SPB should ultimately be judged not as another multilevel converter, but as a modular traction-drive architecture that exchanges semiconductor blocking-voltage stress for converter–machine modularity. Its competitiveness depends on whether that exchange produces a measurable vehicle-level benefit.

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